

Sara Miller

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EDUCATION

BESc. Mechatronic Systems Engineering, University of Western Ontario

Sept 2022 – May 2027

- Dean's list - Achieved 80% or greater average.

WORK EXPERIENCE

Mechanical Design Intern, Honda of Canada Mfg.

May 2025 - Aug 2026

- Mechanical design lead for multiple automation projects, owning design for robotic tooling, manufacturing fixtures, jigs, and production equipment from concept development through fabrication, installation, and validation.
- Created and maintained CATIA V5 part models, large assemblies, detailed drawings, and revision-controlled design packages used by suppliers, manufacturing, maintenance, and installation teams.
- Applied FEA, rigidity calculations, robot reach studies, payload calculations, and interference checks to validate designs and reduce technical risk before implementation.

Mechatronic Systems Engineering Co-op Student, Honda of Canada Mfg.

May 2024 - Aug 2024

- Developed and delivered a course on pneumatic systems: circuit diagrams, and troubleshooting techniques, delivered to interns, technicians and engineers, course is continuing to be taught after my departure.
- Gained foundational experience in Mitsubishi PLC programming, supporting small-scale automation updates and troubleshooting logic-based control systems.
- Developed foundational experience with Cognex machine-vision and vision-guidance systems integrated with robotic automation.

High Ropes Team Lead, Blue Mountain Resort

May 2023 - Aug 2023

- Led a team of 20 high ropes course guides, fostering a collaborative and safety-focused environment
- Acted as the primary point of contact for guest inquiries and concerns, consistently resolving issues to maintain a high level of customer satisfaction
- Maintained up-to-date knowledge of industry safety standards and ensured compliance with all relevant regulations and protocols

Sales Associate, SVP Sports

July 2020 – Jan 2024

- Responsible for end of day cash tallying and balance confirmation
- Managed inventory, including daily receiving and shipping of product orders, trained new employees

SELECTED PROJECTS

HCM Grind Point Robot Automation | Mechanical Design Lead

- Mechanical design lead for \$1.8M project, designed all equipment replacing manual process with automated robot cell, improving quality and reducing downtime.
- Used CATIA V5 to develop robot tooling for automated grinding process, established mounting interfaces, and performed interference and clearance checks across robot poses to confirm access to the work area and avoid
- Designed an automatic tool changer, accounting for tool pickup and drop-off alignment, repeatability, pneumatic connections, payload distribution, service access, and safe retention of the grinding tool.
- Validated the tooling through robot payload calculations, reach studies, design reviews, and engineering analysis, then incorporated feedback from controls, manufacturing, and installation stakeholders.

Miniature C-Arm Fluoroscopy Simulator

- Designed a 3-axis mechatronic system and 3D-modeled structural components for a scaled fluoroscopy simulator.
- Programmed an Arduino-based control system. Integrated accelerometer for arm orientation feedback and system monitoring.
- Integrated accelerometer-based orientation feedback to support potential real-time visualization and future connection to clinical simulation software.
- Designed and 3D-printed a miniature three-axis C-arm fluoroscopy simulator in SOLIDWORKS, creating a safe, portable, and cost-effective alternative for medical training.
- Integrated an Arduino Nano ESP32, stepper motors, rack-and-pinion mechanisms, and wireless serial control to achieve repeatable positioning across three degrees of freedom.

EXTRACURRICULARS

Orientation Week Leader, Engineering SOPH

Feb 2026 - Present

- Mentored first-year engineering students, fostering an inclusive and supportive community through transition programming, group activities, and peer guidance.

Head of Technical Communications, Western Engineering Concrete Canoe Association

Sept 2025 – Aug 2026

- Led the creation and management of technical documentation across multiple sub-teams, ensuring consistent communication of project workflows and producing detailed technical reports.

SKILLS

CAD/Design: CATIA V5, DELMIA V5, SolidWorks, AutoCAD Electrical

Programming/Controls: Java, C, C++, MATLAB/Simulink, Mitsubishi PLC

Engineering Methods & Analysis: GD&T, FEA, DFMA, Pneumatic Circuit Design, Payload Calculations, Rigidity Calculations,

Professional: Technical Communication, Leadership, Project Management, MS Office, Excel